

Pricing and optimization

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Definition: Revenue / Yield Management

Definition: Yield Management

Yield management is the practice of pricing to maximize the amount of revenue received per unit sold.

It is commonly associated with the pricing practices of airlines, hotels, and other sellers of perishable products.

Source: <https://www.youtube.com/watch?v=-B3-IVkUaGQ>

created at <http://www.b2bwhiteboard.com>

Pricing

Important parts of pricing

- Price response (function)
- Price sensitivity
- “Willingness-to-pay”
- Price optimization
- Pricing strategies



Price response: The demand function

The **price response function** assigns to each price level a possible demand

$$d : \mathbb{R}_+^0 \rightarrow \mathbb{R}_+^0, p \mapsto d(p)$$

Here, we assume that the price response function is time-independent.

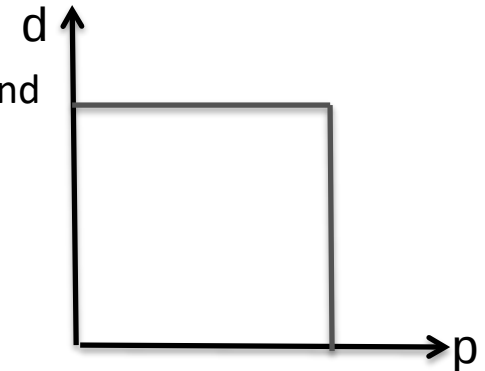
Typically, this is not the case!

Moreover, the price response function is unknown and needs to be estimated

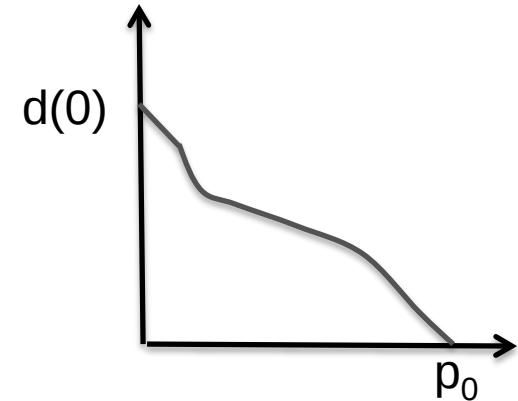
Assumed properties of the price response function

- non-negative
- continuous
- (strictly) monotonously decreasing
- (at least piecewise) differentiable
- (ideally) invertible,
i.e. there is a one-to-one relation between price and demand

Perfect market



Typical price response



Price sensitivity and price elasticity

The price sensitivity describes/measures the slope of the price response:
How does the demand change when the price changes?

Slope

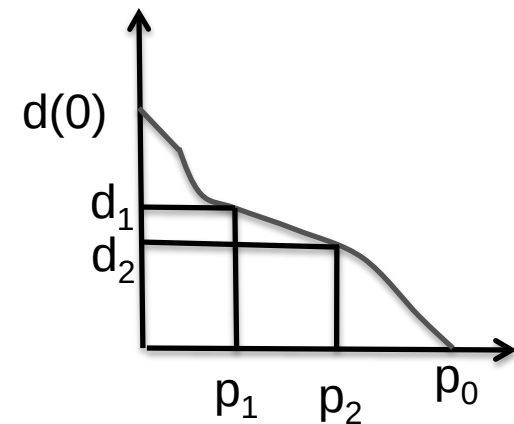
$$\delta : \mathbb{R}_+^0 \times \mathbb{R}_+^0 \rightarrow \mathbb{R}_+^0, (p_1, p_2) \mapsto \frac{d(p_2) - d(p_1)}{p_2 - p_1}$$

Differential quotient vs. Diffence quotient

$$d' : \mathbb{R}_+^0 \rightarrow \mathbb{R}_+^0, p \mapsto \lim_{h \rightarrow 0} \frac{d(p+h) - d(p)}{h} = d'(p)$$

Price elasticity

$$\varepsilon : \mathbb{R}_+^0 \times \mathbb{R}_+^0 \rightarrow \mathbb{R}_+^0, (p_1, p_2) \mapsto -\frac{100[d(p_2) - d(p_1)]p_1}{(p_2 - p_1)d(p_1)}$$



(Theoretical) Willingness to pay

The willingness to pay describes if the customers are willing to pay a price of p . $w : \mathbb{R}_+^0 \rightarrow [0,1], p \mapsto w(p)$

The fraction of the customers that are willing to pay a price between p_1 and p_2 are given by the integral

$$\int_{p_1}^{p_2} w(x) dx$$

Assuming $D = d(0)$, we obtain

- the corresponding demand
- the corresponding willingness-to-pay

$$d(p) = D \int_p^{\infty} w(x) dx$$
$$w(p) = -\frac{d'(p)}{d(0)}$$

Revenue

The achievable revenue is the product of demand times price level.

$$r : \mathbb{R}_+^0 \rightarrow \mathbb{R}_+^0, p \mapsto r(p) = d(p) \cdot p$$

A sufficient criterion for a local maximum of r is given by

$$r'(p) = 0, r''(p) < 0$$

By the product rule, we obtain $r'(p) = d'(p) \cdot p + d(p)$

For a local maximum at a price level p we observe a price elasticity $\varepsilon(p)$

$$\varepsilon(p) := -\frac{d'(p) \cdot p}{d(p)} = 1$$

If the current price elasticity is less than 1, we can increase the revenue by increasing the price.

Pricing strategies

Possible pricing strategies and risks:

- be competitive and offer a price as low as possible
(high sales but most probably low revenue and risk of being sold out quickly)
- stay above a minimum price level: “(cost + x)-pricing”
- monitor and estimate the demand (price response) and learn the potential demand (price response) from the past
- monitor competitors and estimate their price levels and inventory levels
(game theory: competitive game)
- keep your price level (“high quality” product, risk of low sales)
- estimate the current and future demand level, price response, and willingness to pay
- ...